ABDULAZIZ TOBA ABDULAZIZ

AI ENGINEER | LLM ENGINEERING | LOCAL / SELF-HOSTED AI | AGENTS | LLM INFERENCE

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PROFESSIONAL SUMMARY

AI Engineer with software engineering experience since July 2019 and a strong production backend foundation in C#/.NET, APIs, databases, distributed services, and Redis. Progressed from senior backend engineering into applied AI, building self-hosted LLM systems, LangChain/LangGraph agents, voice-agent workflows, summarization, and localized generation features. Current specialization: local LLM inference, model serving, GPU/VRAM sizing, quantization, batching, and latency/throughput benchmarking.

TECHNICAL SKILLS

AI / LLM: Local & Self-Hosted LLMs, AI Agents, LangChain, LangGraph, Hugging Face, Vector Databases, Meeting/Text Summarization, Text & Image Generation

Inference: Model Serving, Quantization, Batching, Continuous Batching, GPU/VRAM Sizing, Benchmarking, TTFT, Tokens/Second, Throughput, Latency, Concurrent Inference, Context/Memory Planning

Backend: C#, .NET 8, ASP.NET Core, REST/Web APIs, Redis, SQL Server, Entity Framework Core, Dapper, Microservices, CQRS, Clean/Hexagonal Architecture, SignalR, gRPC

Additional: Angular, TypeScript, Azure, AWS, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

INVENTO **Jan 2025–Present**

Senior Software Engineer → AI Engineer

- Designed and implemented agentic LLM workflows using LangChain and LangGraph, orchestrating conversation flows and integrating local/self-hosted models with backend services.
- Developed the Voice Agent's LLM integration and conversation workflows, connecting local/self-hosted models with production backend services.
- Built local-model features for meeting/text summarization and localized email generation; integrated Hugging Face, vector databases, Redis, and text/image-generation capabilities into AI workflows.
- Focused on local LLM inference engineering: GPU/VRAM sizing, quantization, batching/continuous batching, concurrent inference, and context-memory tradeoffs; benchmarked TTFT, tokens/second, latency, throughput, and memory utilization.
- Earlier phase: designed scalable .NET/C# APIs and production services, integrating databases and Redis while contributing to backend architecture, performance, and system design.

MANNAI CORPORATION QPSC — QATAR MINISTRY OF ENVIRONMENT & CLIMATE CHANGE (MEC) **Jan 2023–Jan 2025**

Senior Software Engineer | Qatar National Super App / MEC Services

- Developed secure, modular backend services for Qatar's National Super App using .NET 8, C# 12, ASP.NET Core Web API, and Clean Architecture.
- Designed the Digital Falcon Service and integrated CITES to support falcon permits, regulatory compliance, and secure data exchange.
- Led core feature delivery, including the Camping Permit Service with dynamic validations and geolocation; owned production-critical features, architecture decisions, code reviews, and technical design.

APEX SYSTEMS **Jul 2019–Jan 2023**

Senior Software Engineer

- Architected .NET web applications and APIs using CQRS, Hexagonal Architecture, and microservices; implemented Redis caching, SignalR, and Prometheus/Grafana monitoring, managed production deployments, and mentored junior engineers.

EDUCATION

6 October Academy — Bachelor's Degree in Engineering Jun 2018